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USING SIMULATIONS TO UPSKILL EMPLOYEES

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Reskilling workers has become a key challenge for many businesses. Korn Ferry estimates that there will be an 85-million-person talent shortage by 2030.¹ McKinsey likens the challenge to the shift from agricultural work to manufacturing that occurred during the early 20th century. No surprise, then, that training employees tops the agenda in many organizations, according to a LinkedIn report.² The market for global corporate learning and development (L&D) is \$350 billion and increasing.³ Despite this massive industry-wide investment, one study determined that 75% of senior managers at organizations were dissatisfied with their L&D initiatives.⁴ Conversely, Gartner reports that 70% of employees feel they lack the skills required to do their jobs.⁵

There are three systemic reasons for this paradox: 1) an over-reliance on classroom training rather than the on-the-job behavioral practice required by most jobs, 2) a failure to recognize the importance of peer learning and modeling behavior: Most adults learn best by watching how the best of their peers perform key tasks, and 3) a lack of ongoing feedback through relevant performance management processes.

In this article we will examine why the shortfall has occurred, what companies have done to train employees in a more efficient and interactive way, and how technologies can help make L&D processes more effective.

Simulation Training and Management Development

Approaches based on proven learning techniques and new technology can improve the effectiveness and efficiency of skills training. One approach is simulation training: putting people in immersive, true-to-life scenarios where they practice skills acquisition in situations that replicate job conditions. These immersive scenarios often incorporate gaming elements that increase motivation, attention, and learning.

This approach has long been part of skills training in certain sectors. A study of laparoscopic surgery found that surgeons who trained using simulations had a 29% increase in speed and a nine times lower likelihood of experiencing a stall during surgery, and were five times less likely to cause patient injury.⁶ Brigham and Women's Hospital now uses the technique for multiple procedures via its STRATUS Center for Medical Simulations in its Surgical Leadership Program. Pilots use flight simulators because

their training requires reinforcement, periodic upgrading in new conditions, and the ongoing motivation that is a byproduct of a good developmental process. Or, as stated more bluntly in the “7Ps” of the U.S. Air Force, “Proper prior planning prevents pitifully poor performance.”⁷

Simulation training can help improve core skills in two critical components of management development: 1) decision-making and collaboration and 2) change management.

1. Decision-Making and Collaboration

A global digital agency sought to improve managing directors’ (MDs) decision-making in three areas: when to collaborate with other MDs, how to price projects, and staffing project teams. In most organizations, these skills are seen as something you only learn through experience and, if you’re lucky, with mentors. But it’s no coincidence that experiential learning is called the “school of hard knocks.” In most firms, on-the-job learning is a euphemism for an ad hoc process, not a deliberate practice.

The agency had training programs in place, but a game-based solution called Financial Fitness Simulator enabled more than 300 MDs to work through scenarios involving discovery conversations with clients, enlisting support from other service lines, and managing projects while building a pipeline of prospects. After this initiative, 88% of the MDs reported a much better understanding of their responsibilities and improved decision-making skills; 90% said their ability to collaborate with other lines of business had also improved. These are compelling results compared to most training initiatives.

The simulation scenarios involved narratives, or plotlines depicting common work situations. This is a key element in promoting durable learning. As one summary of the research puts it: “Narrative provides not only meaning but also a mental framework for imbuing future experiences and information with meaning, in effect shaping new memories to fit constructs of the world and ourselves.”⁸ Learning theorists call this “active retrieval.” When people must respond to diverse circumstances, learning involves the ability to retrieve a relevant model or scenario, and this ability is extended with each iteration.

At the digital agency, the participants moved through scenarios representing different stages of the fiscal year, types of prospects, and client projects. They had to balance executing ongoing projects, adding relevant prospects, coordinating required firm resources, and prioritizing decision criteria based on trade-offs like focusing resources on a smaller repeat customer or a larger but riskier deal. Scenarios occurred within an immersive environment involving dialogue and online interaction with other MDs, new and experienced, which gave participants access to collective knowledge about trade-offs and likely consequences.

The simulations outperformed any previous classroom training. MDs engaged with the simulation six times on average, because they could access this on-demand tool when it fit into their schedules, unlike scheduled classroom instruction, and with feedback each time they participated (as opposed to at the end of a month-long seminar). With responses to multiple scenarios captured online, participants develop situational awareness and identify areas for improvement. This, in turn, creates more productive learning as participants bring specific questions to subsequent sessions. Across fields, research shows that people become high performers via deliberate practice of specific skills in safe environments.⁹ “Role plays” are part of many training initiatives but are typically one-off sessions with a manager, not easily scalable, and hampered by fear of judgment from one’s superior. The Financial Fitness Simulator involved practice in a context where the stakes of initial failure did not include career risks.

The simulations also incorporated gaming principles. Participants gained or lost points based on their decisions and with visual and audio reflections of success or error (e.g., green points and celebratory noises for correct answers; red points with a negative-sounding noise for wrong answers), while leaderboards displayed the highest earners (unless they chose to remain anonymous), allowing learners

to compete for global bragging rights with peers and establish virtual but real social ties. These gamified elements are proven techniques for increasing engagement and motivation. Numerous participants echoed the comments of one MD: “I like the interactive nature [and] ability to go back and try it again, thus learning the ‘why’ behind the right answer.”

2. Change Management

Dealing with change was a challenge facing Merck in upskilling 650 frontline sales managers in its health care business. Speed was important: In this business, a patent clock is ticking, so time and money are closely correlated. More generally, studies indicate that “I’m too busy” is the most frequently cited barrier to managerial participation in training.¹⁰ The ability to deal with diverse buying and regulatory circumstances was important, too: The initiative involved 40 virtual-reality simulations for managers across 65 countries, translated into seven languages with attention to cultural nuances.

Topics ranged from goal setting and problem-solving to coaching and performance reviews. In the simulations, managers met a range of reps with different personalities, capabilities, and challenges. Some are demotivated, while others have difficulties getting client meetings, managing sales conversations, or handling scenarios like a breach of client data. The simulations are played on managers’ laptops, and each takes place in a realistic setting like a doctor’s office, conference room, office environment, or on a train. The manager must select the most appropriate approach to the issue in that setting, weighing the information available, and then receives feedback about best practices in that context. The simulation includes more than 200 minigames at different levels that build managers’ knowledge of specific topics, as well as realistic dialogues with avatars representing internal and external stakeholders. The dialogues change based on how managers respond to each scenario, which determines their unique journey through the simulation—a contrast to the lowest-common-denominator approach in much traditional leadership training and itself an important motivator.

Chetak Buaria, Vice President of Global Commercial Operations for Merck’s health care business, described the initiative as a “phenomenal success which significantly improved the effectiveness of Merck’s sales leadership team.”

The reasons for that success are relevant to other upskilling situations. Compared to training in a classroom setting, Buaria says this new approach “reduced the training time by 70%, saving over 2,000 working days of frontline managers’ time.” That’s important in professional development, especially in functions like sales. An effective learning process should not take net time away from productive selling activities.

The approach also motivated participant engagement. Some managers devoted dozens of additional hours of their personal time to the simulations—a remarkable feat for busy sales managers with quotas, one that is not typically observed in e-learning initiatives. A simulation approach recognizes a key facet of adult learning: Professionals tend to pay attention to information when they need it, not weeks earlier or later in a training seminar. In sales, for example, this is typically when preparing for a sales call or coaching session or during those conversations. The simulations allow for just-in-time learning in contexts and time frames connected to actual job tasks.

Finally, upskilling typically involves frequent updating of content to reflect changing circumstances. Sales training is an example. Training here must be a process, not an event. At any point in time, sales reps and managers are dealing with multiple accounts and a changing array of people and buying criteria. And the learning must recognize differences across countries and categories. Compared to other training initiatives, a virtual simulation can be more easily modified and in formats conducive to the manager accessing and practicing skills when that practice has the most impact.

Simulation opportunities are expanding. The so-called “metaverse” may or may not deliver on its current hype, but the investments being made by companies in virtual reality increase training and simulation options. Meanwhile, outside a classroom, which is where most adults earn a living, learning is itself a learned behavior. These examples, grounded in sound theory and supported by increasingly accessible and flexible technologies, indicate how companies can take advantage of these developments and address a major societal workforce issue while getting more from their training investments.

TAKEAWAYS

Challenges around reskilling and upskilling stem from an overreliance on classroom training, a failure to recognize the importance of peer learning, and a lack of ongoing feedback. Simulation training for management development can help address these gaps.

- # Simulations involve putting people in immersive, true-to-life scenarios where they practice skills acquisition in situations that replicate job conditions.
- # These immersive scenarios often incorporate gaming elements that increase motivation, attention, and learning.
- # Simulations can help improve core skills decision-making, collaboration, and change management.
- # The opportunities for simulations are expanding. Investments made by companies in virtual reality increase training and simulation options.

NOTES

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